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الحياة الطيبة

Software Requirements & UML Specification

Client review draft | Version 0.1 | 27 September 2026

A single-topic MVP for children aged 8-12 and their mothers, with independent learning on separate devices and an extensible content-management workflow.

Review status	Purpose
Draft for review	Confirm scope and behavioral requirements before UI design and detailed implementation.
Confirmed and proposed items	Confirmed product direction is distinguished from proposed mechanisms and unresolved decisions.

Includes the SRS, eight detailed use cases, logical architecture, eight UML diagrams, and requirements-to-acceptance traceability.

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How to read this document

Start with the executive brief and decision register. Review the SRS and use cases before treating the models as an implementation baseline. The UML section uses larger pages where necessary to keep technical labels readable.

Status notation

D - derived: operationalizes a confirmed product decision. **P - proposed:** mechanism or scope awaits agreement. **Must / Should:** priority within the proposed MVP. Normative wording makes a draft requirement testable; it does not mean it has already been approved.

01 Executive brief

Product direction

Hayah Tayeba helps children aged 8-12 understand a value, explore responses through a story, and try a small action in everyday life. Mothers have both their own learning journey and optional ways to support their children. Each uses a separate device. The mobile client will use Expo React Native.

The first journey is «لما أزعل، أتصرف إزاي؟» inside «باب حسن الخلق». The setting is lightly fantastical, while situations and dialogue remain familiar. The core message is that a child can acknowledge feeling upset and learn how to respond; completion is not a judgment of character or spiritual growth.

MVP boundary

- One complete child journey, corresponding mother content, and an optional shared activity.
- Independent child progression after the proposed account setup.
- Saved learning progress and recovery from ordinary connection interruptions.
- An admin workflow for adding content using supported templates.
- Explicit content review, publication, revisions, and withdrawal.

Decisions required before implementation

- Adult sign-in, initial child setup, and device pairing.
- Exactly what the mother may see about the child's activity.
- Content authorship, review ownership, and the mother's first lesson.
- Launch platforms, distribution countries, and connectivity expectations.
- Data-retention policy, operating ownership, and release quality targets.

Delivery status

This document specifies a proposed solution. Application development, final curriculum production, religious-source verification, and product acceptance testing have not been completed. The supplied presentation and expanded project notes are conceptual inputs rather than an approved finished curriculum.

Review approach

Agree the behavioral scope first, then select backend technologies and physical data/API designs. UI prototyping can still reveal issues that require revisions to the specification. No price, delivery date, hosting provider, or launch commitment is established by this draft.

02 Decisions and open questions

Confirmed product direction

ID	Confirmed decision
C-01	Children are aged 8-12.
C-02	The mother has a personal journey and a role supporting her child.
C-03	Children can start and complete their experience independently; mother participation is optional.
C-04	MVP begins with «لما أزعل، أتصرف إزاي؟» inside «باب حسن الخلق».
C-05	Light fantasy setting with everyday situations and dialogue.
C-06	Mobile client uses Expo React Native.
C-07	Mother and child use separate devices.
C-08	Admins can periodically add new content.
C-09	Start with one topic in the MVP.
C-10	The existing project deliverable is the presentation.
C-11	Hold UI design and prepare SRS/UML first.

Proposed defaults - awaiting confirmation

ID	Proposal	Reason and limit
P-01	Mother authenticates; child uses a linked profile and a device session without a child email or phone.	Provides separate-device access without requiring child contact details. Identity provider is undecided.
P-02	Temporary, single-use pairing code with explicit mother confirmation of the pending device.	Makes linking understandable while reducing accidental linking. Detailed timing/rate limits are provisional.
P-03	One responsible mother account per child in MVP; one mother may manage multiple children.	Bounds ownership and authorization. Additional guardians and ownership transfers are deferred.
P-04	Mother's overview exposes topics and completion, not individual choices or personal reflections.	Separates support from detailed behavioral surveillance. This visibility boundary requires confirmation.
P-05	Do not collect or persist free-text reflections in the MVP.	Avoids introducing sensitive content and moderation needs before requirements exist. Optional fixed-choice reflection stays on-device and is excluded from reports/analytics.

ID	Proposal	Reason and limit
P-06	Template-driven content: illustrated scenes, optional narration/video, choices with feedback, real-life activities, optional reflection.	Admins can add content using existing renderers; novel game mechanics still need development.
P-07	Admin console is a separate desktop web application.	Easier authoring and preview. Framework is undecided.
P-08	One admin role initially; review and publication are explicit actions, possibly by the same authorized person.	No unrequested multi-role approval bureaucracy. Independent editorial review can be added if the team requires it.
P-09	Immutable published content revisions; attempts bind to a revision.	Protects in-progress sessions and historical completion. Withdrawals need explicit handling.
P-10	Online content access, with durable local progress and a retry queue; no full offline catalog/download manager.	Handles ordinary interruptions without claiming complete offline operation.
P-11	No compulsory streaks, leaderboards, spirituality scores, or mother sign-off to advance.	Completion measures learning participation, not moral or spiritual worth.
P-12	Arabic RTL first; child copy can use accessible Egyptian Arabic. Audio/text alternatives for essential instructions.	Matches supplied material; dialect and narration budget still need confirmation.
P-13	One backend deployment with modular responsibilities, transactional data, and separate media storage.	A starting architecture, not a selected provider or mandatory infrastructure stack.
P-14	First journey uses illustrated panels and choices before investing in full animation or custom games.	Tests the end-to-end experience with fewer production dependencies.
P-15	One active child learning-device session per profile; replacement revokes the prior session.	Simplifies cursor synchronization; multiple children can still use separate devices. Simultaneous devices for the same child require a later conflict policy.

Open decisions

The timing column identifies when each decision is needed for the next delivery stage.

ID	Question	Needed before	Working treatment
OQ-01	Accept linked child profiles and mother-approved device pairing? What adult sign-in method?	Auth implementation	P-01/P-02 modeled provisionally.
OQ-02	What exactly may a mother see about her child's activity?	Data schema and API baseline	P-04/P-05 minimize disclosure.
OQ-03	Who authors and reviews educational/religious material and media usage rights?	Production content publication	Review fields exist; no reviewer authority is invented.
OQ-04	Is Android first, iOS first, or both at launch? Minimum OS/device/browser support?	Build and QA plan	Expo chosen; platform release scope remains open.

ID	Question	Needed before	Working treatment
OQ-05	Must complete journeys work offline, or is interruption recovery enough?	Storage/media design	P-10 is the draft scope.
OQ-06	Is the first release private pilot or public distribution, and in which countries?	Launch/privacy policy	No legal compliance determination made.
OQ-07	What is the mother's actual first lesson and personal practice?	Curriculum acceptance	A parallel anger-response lesson is an example only.
OQ-08	Separate content adaptations for 8-9 and 10-12, or one initial journey tested across both?	Content production	One initial version proposed; usability reviewed across ages.
OQ-09	Is initial adult setup acceptable before a child can start independently?	Onboarding baseline	Independence during learning is confirmed; zero-setup guest mode is not.
OQ-10	Monetization, budget, expected pilot size, media volume, and availability target?	Hosting and operating plan	No ads, payments, or subscription system included in draft MVP.
OQ-11	Account deletion, retention periods, backup retention, consent notices, and support owner?	Public release	Draft deletion behavior is specified; periods remain TBD.
OQ-12	How should editorial safety withdrawals affect unfinished sessions?	Content lifecycle implementation	Stop withdrawn content, preserve records, offer another available route.

Items not yet adopted as requirements

“بلدة طيبة” as a world name; exact palette; a particular mascot; full avatar customization; a fixed lesson duration; collectible artwork; push notifications; a streak system; automatic content personalization; any AI tutor; any named backend/database/cloud vendor. Exploratory visual concepts do not constitute approved product requirements.

Tradeoffs to revisit

- A curated route is easier to author than arbitrary branching. The initial editor should allow bounded choices and explicit feedback, not a general game engine.
- Detailed parent visibility can support intervention, but can also discourage honest reflection. Resolve the boundary explicitly before storing answers.
- Dynamic content reduces app releases only for supported templates and schema versions. New client capabilities need a client update.
- One shared age range lowers production cost, but usability testing may show that reading level, pacing, and visual style need variants.
- Pausing UI design is a sequencing choice. A small flow prototype may still be needed to validate pairing and child comprehension before finalizing requirements.

03 Software requirements specification

1. Purpose and interpretation

Specify the behavior of the first usable product and its content-management capability before visual design and backend technology selection. This is a tailored SRS, not a claim of conformance to a particular standard.

D (derived) means a requirement directly operationalizes a confirmed product decision. **P (proposed)** means its mechanism or scope needs product approval. Both remain part of a draft. “Shall” makes the proposed baseline testable; it does not imply approval. Sources C-xx and P-xx are defined in Section 02.

Priority: **Must** = required for the proposed MVP; **Should** = desirable but negotiable. Items explicitly deferred are not MVP commitments.

2. Product purpose, scope, and success

Help children aged 8-12 explore a value, practice a response, and try a small action in everyday life. Mothers pursue their own learning and optionally support their children. The first topic is «حسن الخلق» and the first journey is «إزاي؟ لما أزعل، أتصرف».

The product shall represent learning participation and practice, not certify spiritual growth, moral worth, or emotional health. Completion is not evidence that a child has changed behavior outside the app.

Proposed MVP includes

- Separate mother and child experiences using Expo React Native.
- A proposed family-linking and device-access flow.
- One complete child journey; one corresponding mother lesson/practice whose final content is still pending; an optional shared activity.
- A lightly fantastical, tappable world with everyday situations.
- Supported content templates and an admin authoring/publication workflow.
- Persisted progress, resumption, duplicate-safe synchronization, and basic recovery states.
- Authorization, accessibility, editorial provenance, and operational controls appropriate to this scope.

Deferred

Custom game builder, free-roaming 3D world, generative AI/chat, social messaging, public child profiles, user uploads, competitive rankings, payments/ads, full offline media library, teacher/school roles, multiple guardians, push campaigns, and automated adaptive learning.

Acceptance outcomes

- A linked child can finish the available journey without adult approval during learning.
- A mother can complete her own content independently of the child's progress.
- An admin can publish a second journey using supported templates without changing the client binary.
- Existing completion and in-progress sessions remain interpretable after a routine content revision.
- A network interruption does not silently erase acknowledged progress or cause duplicate completion.

Pilot engagement and learning effectiveness need observation and a separate evaluation plan. This SRS sets no invented adoption or efficacy targets.

3. Actors and boundaries

Actor	Allowed capability	Boundary
Child	Access their published journey, make choices, complete activities, resume progress.	No admin controls, adult account settings, other children's records, or public profile.
Mother	Manage her account and linked children, complete her own journey, view the permitted child summary, optionally participate.	No access to unrelated families; detailed child reflections excluded under P-04/P-05.
Admin	Author, preview, review, publish, revise, and withdraw content.	Content role alone grants no access to private learner records.

Authentication and media services are external technical dependencies, not additional human roles. A support/operator role is not specified for MVP; production ownership must be resolved before launch.

4. Functional requirements

Accounts, linking, and access

FR-01 | P / Must / P-01

The system shall authenticate mothers and admins and enforce capabilities on the server.

Acceptance: an unauthenticated request cannot retrieve family data; a mother or child session cannot publish content even if the request is crafted directly. Provider and recovery mechanism remain OQ-01.

FR-02 | P / Must / P-01, P-03

A mother shall create a linked child profile with nickname, age in the supported range, and a preset avatar; child email/phone are not required.

Acceptance: out-of-range ages receive a clear message; the profile is associated with exactly one mother account in this MVP. Exact birth date is not collected.

FR-03 | P / Must / P-02

A mother shall initiate a temporary, single-use pairing challenge; the child device submits it; the mother explicitly confirms that pending device before a child session is issued.

Acceptance: expiry, cancellation, reuse, or confirmation by a different mother fails; simultaneous redemption cannot create two authorized sessions from one challenge. Pairing values are not logged.

FR-04 | P / Must / P-01, P-02, P-15

A mother shall view and revoke a linked child's device session and pair a replacement without deleting progress.

Acceptance: subsequent protected server requests from the revoked session are rejected; replacement pairing revokes the previous active learning session; replacement device receives server-confirmed progress. Unsynced progress on a lost device cannot be promised recoverable.

FR-05 | D / Must / C-02, C-07

The system shall maintain separate mother and child learning contexts across their devices.

Acceptance: the child's session cannot switch to the mother's private account; the mother's learning completion does not mark a child's step complete, or vice versa.

Child learning and progress

FR-06 | D / Must / C-04, C-05, C-09

The child shall see the published MVP door and journey in a lightly fantastical setting, with a clear start/resume action.

Acceptance: the MVP can be used with only one door and one journey; unpublished content is not visible to child sessions.

FR-07 | P / Must / P-06, P-09

The client shall fetch a validated, versioned content manifest and render only supported schema/template versions.

Acceptance: unsupported content does not crash the client or advance progress; a suitable update/unavailable message appears. Existing supported content remains accessible.

FR-08 | P / Must / P-06, P-14

Story steps shall display illustrated scenes, short dialogue, and optional narration with accessible playback controls.

Acceptance: essential meaning remains available as text when audio is muted/unavailable; navigation does not depend on audio completing.

FR-09 | P / Must / P-06, P-11

Choice steps shall provide authored feedback or a consequence and permit another attempt.

Acceptance: each choice has a valid destination or feedback; no score of faith/character is inferred; trying another response does not erase existing completion. A required choice step completes after exploring at least one choice and its feedback, not after guessing an approved answer.

FR-10 | D / Must / C-03

A child shall be able to finish required learning steps without mother attendance, permission, or a shared activity.

Acceptance: decline/skip of every optional shared step still permits journey completion.

FR-11 | P / Must / P-06, P-11

Real-life activities and reflections shall support deferral or skipping.

Acceptance: “أجربها بعدين” and “مش حابب أجاوب” do not block the next step or completion. No proof of a real-world event is required.

FR-12 | P / Must / P-09, P-10

The system shall persist the last checkpoint and required-step completion for each attempt, queue unsent progress durably, and retry safely.

Acceptance: force-closing after a local save resumes at that checkpoint on the same device; duplicate event submission has one effect; the UI distinguishes pending synchronization from confirmed synchronization.

FR-13 | P / Must / P-09, P-11

A journey shall complete when all required digital steps in its bound revision meet their completion rules.

Acceptance: optional real-life/shared/reflection steps do not gate completion; a completion acknowledgment occurs once per attempt; a replay does not retract historical completion.

Mother experience

FR-14 | D / Must / C-02

The mother shall have an independent content path with its own saved progress and a personal practice.

Acceptance: it works before, during, or after the child's journey. The first lesson must be authored and reviewed before release; it is not yet supplied.

FR-15 | P / Must / P-04, P-05

A mother shall see only the permitted summary for each linked child: topics accessed, required-step/journey completion, and authored support suggestions.

Acceptance: server responses exclude raw choice histories, optional reflection answers, spiritual scores, and unrelated-family data. Pending child-device progress appears only after successful synchronization.

FR-16 | D / Must / C-02, C-03

The system shall offer optional shared activities alongside solo routes.

Acceptance: neither device must be online simultaneously and no real-time parent/child chat or co-play session is required.

Admin content management

FR-17 | D / Must / C-08

An authorized admin shall add/edit doors and journeys and order their content.

Acceptance: a second supported journey can be created, published, and retrieved by an existing client without a code change.

FR-18 | P / Must / P-06

The editor shall support a bounded template catalog: illustrated story, media, choice/feedback, real-life activity, optional reflection, and shared activity.

Acceptance: every step has a stable ID, audience, type, required/optional flag, and valid template data; unsupported types are rejected by validation.

FR-19 | P / Must / P-06

Admins shall associate images/audio/video with usage-rights notes, meaningful accessibility text/captions, and processing status.

Acceptance: unsafe or unsupported file types, unfinished uploads, and missing required assets cannot be published; upload alone never makes media public.

FR-20 | P / Must / P-08

Admins shall preview each audience's journey and record an editorial review before publication.

Acceptance: unresolved references, unreachable required steps, missing essential alternatives, unreviewed religious quotations, and unsupported schemas block publication with specific errors. A second distinct reviewer is not required by this draft.

FR-21 | P / Must / P-09

Publishing shall atomically expose an immutable revision and update the catalog pointer.

Acceptance: editing live content creates a draft; a failed publish exposes no partial journey; concurrent stale edits/publishes are rejected rather than overwriting newer work.

FR-22 | P / Must / P-09

Routine revisions shall preserve attempts bound to older available revisions and historical completion.

Acceptance: new attempts use the current revision; existing attempts resume their revision unless withdrawn; a future optional second journey does not undo completion of the first.

FR-23 | P / Must / P-09

An admin shall withdraw a published revision with a reason and explicit confirmation.

Acceptance: it disappears from new-start catalog results; the next online access check blocks further delivery for bound attempts and offers an explanation; historical progress remains. The app cannot guarantee immediate revocation of already-delivered offline bytes.

FR-24 | P / Must / P-08, P-09

The system shall audit authoring lifecycle changes, review, publish, and withdrawal with actor, timestamp, revision, action, and reason where needed.

Acceptance: failed actions are distinguishable from successful publication; secrets and child answers are absent from audit payloads.

Recovery and account lifecycle

FR-25 | P / Must / P-10

The client shall recover gracefully from network and media failures.

Acceptance: a visible retry and saved checkpoint remain; it does not claim that unfetched media works offline; a failed optional media load does not permanently trap a user.

FR-26 | P / Must / P-05

Optional fixed-choice reflections shall be skippable and local to the active profile; no free-text reflection is collected.

Acceptance: reflection content is absent from backend requests, parent summaries, telemetry, crash logs, and admin tools; clearing a profile/session clears its local reflection data.

FR-27 | P / Must / OQ-11

Mothers shall be able to request deletion of their account or a linked child profile with scope explained and confirmation required.

Acceptance: a confirmed request immediately revokes relevant sessions and schedules removal/anonymization of associated learner data; admin content authored by another account remains. Retention, completion deadline, and backup purge policy must be agreed before release.

5. Business rules

ID	Rule
BR-01	Optional mother participation is never an unlock condition. Account setup/pairing is a separate unresolved onboarding choice.
BR-02	Anger/sadness may be acknowledged without labeling the child's character. Completion reflects interaction with learning material.
BR-03	New content uses supported templates and schemas. A new engine or game mechanic may require a new client release.

ID	Rule
BR-04	Published revisions are immutable. A draft can be rejected for review without affecting the currently live revision.
BR-05	A completion record identifies learner, journey, and content revision. It remains historical after revision or withdrawal, except deliberate account erasure.
BR-06	Required digital steps must be reachable. Optional/shared/reflection/real-world steps cannot be prerequisites for required progression.
BR-07	For MVP, each child belongs to one mother account. Content admins do not inherit access to learner data.
BR-08	A device credential authorizes a specific child profile, not every child in a family.
BR-09	A withdrawn revision cannot be resumed after the client learns it was withdrawn. Historical records are retained with an unavailable-content explanation.
BR-10	Resume cursor and completion are different: revisiting an earlier scene can change the cursor but cannot undo completed steps.
BR-11	Repeating a request with the same event identifier and payload returns the same outcome; reusing that identifier with a different payload is rejected.

6. Data requirements and visibility

Conceptual entities are modeled in UML. Physical tables and a database vendor are not selected.

Data	Minimum content	Visibility / lifecycle
Mother account	Internal ID, auth-provider reference, preferences	Self and authorized account operations; no public directory.
Child profile	Internal ID, owner ID, nickname, age or age band, preset avatar	Child and owning mother within allowed views; no exact DOB/contact info required.
Pairing challenge /request	Non-plaintext verifier, expiry, pending-device reference, state	Scoped authorization only; single use, no token logging.
Device session	Session identifier, child scope, expiry/revocation metadata	Mother manages own children's sessions; tokens kept in device secure storage.
Door / journey	Stable ID, audience metadata, order, current published revision	Admin drafts private; published eligible metadata to learners.
Content revision / step / choice	Immutable revision ID, schema version, ordered step graph, assets, per-step rules	Readable to authorized learners only when available.
Attempt / checkpoint / completion	Learner, revision, cursor, completed step IDs, sequence, sync metadata	Own learner; mother sees a reduced child summary. No raw answer history stored as progress.
Reflection selection	Optional fixed-choice local value	Device-only under P-05; cleared with profile/session removal, no sync/reporting.

Data	Minimum content	Visibility / lifecycle
Editorial review / audit event	Reviewer/admin, revision, action, time, source/rights notes	Authorized admin operations. Not a general learner activity log.

Server progress accepts only fields needed for completion and resumption. It must not use a permissive analytics/event payload that accidentally stores child choices or reflection text. Media access must follow the same draft/published/withdrawn permissions as metadata.

7. Nonfunctional requirements - proposed targets

All numerical targets below are design proposals, not observed performance. Final test hardware, network profile, pilot concurrency, and deployment region must be recorded before baselining.

NFR-01 | Authorization

Enforce object-level ownership and role checks on every protected operation, including media preview and parent summaries. Verify with unrelated-family and wrong-role requests for each endpoint family.

NFR-02 | Sessions and pairing

Use encrypted transport and secure device credential storage. Pairing challenge expires after 10 minutes; proposed limit is 5 failed submissions per device context/IP window of 10 minutes, with additional server abuse controls. Validate atomically and rate-limit; do not rely on client checks. Exact policy is adjustable after threat review.

NFR-03 | Privacy

No ads/tracking SDKs or sensitive-answer telemetry in this MVP proposal. Review network traces and logs for pairing tokens, reflections, child contact data, and raw answers. Decide retention and account-erasure deadlines before release.

NFR-04 | Reliability

Local progress writes survive app restart once acknowledged locally. Server acknowledgment occurs only after durable commit. Retry tests shall prove no duplicate completion or backward completion regression. Explicitly label unsynced state.

NFR-05 | Interaction performance

Proposed target: local navigation/choice feedback starts within 200 ms at p95 on the agreed representative device. Media startup is measured separately. Do not block navigation on analytics.

NFR-06 | Service performance

Proposed target: metadata/progress APIs complete within 1 second at p95 at an agreed pilot load, excluding media transfer. Cold launches and large assets require their own measured budget. No capacity claim until load/device/network assumptions are agreed.

NFR-07 | Accessibility / language

RTL layouts, text/audio alternatives, captions where essential, non-color-only feedback, reduced-motion support, and accessible control names. Verify key journeys with screen readers and 200% text scaling without loss of actions. No standards-compliance certification is claimed.

NFR-08 | Compatibility

Expo React Native is fixed. OS versions, Android/iOS launch coverage, admin browser support, and device matrix remain OQ-04. A tested matrix must exist before release.

NFR-09 | Content compatibility

Unknown schemas/templates fail safely; publication validates supported client capability requirements. Test a currently supported client against a future unsupported revision without a crash or corrupt progress.

NFR-10 | Operations

Capture errors, sync failures, and publication failures with correlation IDs and no sensitive payloads. Test restore from a backup in a nonproduction environment; backup frequency, recovery objectives, retention, and operator ownership must be set before launch.

NFR-11 | Editorial quality

Every published journey has documented source/rights review and an age-appropriateness review by the project's appointed owner. Final religious text is reviewed before publication; this SRS supplies no approval.

NFR-12 | Child usability

Before release, observe children across the 8-12 range completing the key flow without coaching after setup. Record confusion, reading/audio needs, and failure to find retry/exit actions. Sample size and success threshold remain to be agreed; no untested claim that one design fits all ages.

8. External interfaces

- **Mother/child mobile app:** Expo React Native. Specific screens are design work to follow, not mandated pixel layouts.
- **Admin client:** proposed desktop web app. Writes through authenticated content operations, not direct database access.
- **Identity boundary:** authenticate adults, issue scoped child sessions, revoke credentials, and recover adult access. Provider and login method remain open.
- **Content interface:** paginated/ordered catalog, revision manifest, capability metadata, and authorized media references.
- **Progress interface:** scoped attempt creation, version-bound checkpoint events with idempotency keys, and confirmed-summary reads.
- **Media pipeline:** uploads, validation/processing status, published-delivery references, and withdrawal checks. Provider, codecs, limits, and signed-URL policy are implementation decisions.

No API route names, database engine, hosting platform, or microservice topology is contracted by this document.

9. Failure and change policy

- Expired pairing: regenerate; never link by nickname or accept a guessed identity.
- Interrupted learning: resume locally when required cached data exists; otherwise preserve checkpoint and request connectivity.
- Routine content update: keep current attempt on its version; new attempts use the new version.
- Content withdrawal: stop subsequent authorized delivery; keep an explanatory historical record; do not claim cached bytes can be remotely erased instantly.
- Stale progress submission: merge completion monotonically, validate cursor ordering, and return authoritative state. Device replacement starts from server state.
- Failed publish: keep prior live version and return actionable validation errors.
- Account erasure: stop access immediately; apply the agreed erasure/backup policy without promising a period not yet defined.

10. Release gate and outstanding decisions

Use Section 07 for acceptance scenarios. Before a public release, resolve account setup/recovery, mother visibility, launch platforms/countries, content ownership, data retention/erasure, operational ownership, and the test/device/load matrix. Content production and review are release dependencies, not assumed completed work.

SRS approval, implementation verification, and educational validation are three different milestones. None is implied by creation of this draft.

04 Detailed use cases

UC-01 - Establish family access on separate devices

Actors: Mother, child. **Requirements:** FR-01-FR-05. **Assumptions:** P-01/P-02/P-03; initial adult setup is acceptable, pending OQ-09.

Preconditions: Mother can authenticate; service is reachable. A device does not gain child access merely by choosing “child” on a welcome screen.

Main flow:

1. Mother signs in, reads the proposed profile/linking explanation, and creates a child profile.
2. Service binds the profile to her account; she requests a pairing challenge.
3. Child enters that challenge on their separate device.
4. Service checks validity and reserves a single pending redemption atomically. It does not yet issue a child session.
5. Mother sees the pending request and confirms the device she is pairing.
6. Service checks ownership and challenge expiry again, atomically consumes the challenge, and issues a child-scoped session to the pending device through its protected pending-request credential.
7. Child device stores the session securely and loads its own published catalog and progress.

Alternatives/failures: invalid/expired challenge to retry or regenerate; wrong device to reject/cancel and regenerate; simultaneous redemptions to only one pending reservation; mother does not confirm before expiry to no child session; interrupted completion response to the same pending device can retry and retrieve the same authorization result, not mint an additional session.

Postcondition: authenticated child device is linked to exactly one profile. Mother credentials are not copied to it. No learning step requires further mother confirmation.

Unresolved: adult login/recovery provider, explicit notices, final abuse policy, whether a guest preview should exist.

UC-02 - Explore the first child journey

Actor: Child. **Requirements:** FR-06-FR-11. **Preconditions:** authorized session, published supported content available.

1. Child enters the active door and selects «لما أزعل، أتصرف إزاي؟».
2. Service selects an available revision and creates/resumes a bound attempt.
3. Child views an illustrated scene; narration is optional and core meaning is readable.
4. Child explores a choice and its authored consequence.
5. Child may try a different response and continues after the step's participation criterion is satisfied.
6. Child sees a real-life activity and may defer it; shared participation and reflection are optional.

Alternatives/failures: unsupported schema to explanatory state with safe return; missing required media to retry without losing checkpoint; optional media failure to text alternative; distressing/repeated-mistreatment scenario to authored trusted-adult guidance, not a diagnostic or emergency service.

Postcondition: the child has a meaningful next action and saved learning checkpoint. No moral score is generated.

Example content is provisional: exclusion from a game, expressing sadness, pausing before responding, and seeking help. Final wording, religious evidence, pacing, and graphics require content review.

UC-03 - Save, resume, and complete learning

Actors: Child or mother learner. **Requirements:** FR-12, FR-13, FR-22, FR-25, FR-26.

1. App records the current checkpoint and completion evidence locally using a unique event ID and attempt revision.
2. App marks unsynced progress as pending and submits allowed progress fields to the service.
3. Service checks learner scope, revision availability, step identity, event uniqueness, and sequencing.
4. Service atomically records the event and merges completed steps without regression.
5. Service determines completion from the bound revision's required-step rules and returns confirmed state.
6. App acknowledges synchronization and can resume at the saved checkpoint on later entry.

Alternatives/failures: duplicate request to same result; same event ID/different payload to reject; connection loss to durable queue; force-close to restore local checkpoint; stale cursor event to retain newer cursor; a deliberate later replay of an earlier scene to allow newer cursor while retaining completion; withdrawn revision to stop new content access, retain historical records, show explanation.

Postcondition: progress is either explicitly pending locally or committed on the server. A second device cannot see progress that has never reached the server.

Privacy: choice feedback is processed from authored content; raw choice histories and optional reflection answers are not progress payloads. Completion evidence contains step IDs/statuses, not free-form answers. The server is not certifying real-world behavior.

UC-04 - Complete the mother's personal learning

Actor: Mother. **Requirements:** FR-05, FR-14.

Preconditions: authenticated mother, reviewed published mother content.

1. Mother opens her personal journey, separately from the child overview.
2. She reads/listens to the lesson, considers the suggested personal practice, and completes its required digital steps.
3. Her progress is saved using her own learner identity.
4. Optional reflection can be skipped and remains device-local under the draft policy.

Alternative: child has not started, has completed, or is unavailable to mother's learning is still accessible.

Postcondition: mother completion is independent of every child's attempt.

UC-05 - Support a child optionally

Actor: Mother; child participates only if desired. **Requirements:** FR-10, FR-15, FR-16.

1. Mother chooses one of her linked children.
2. Service returns topics and server-confirmed completion only, through a summary projection.
3. Mother opens an authored support suggestion or shared activity.
4. They may discuss it outside the app; no simultaneous device connection or chat is required.

Alternatives: no child progress yet to helpful empty state; child progress pending on another device to last confirmed state, not invented freshness; activity declined to child can use the solo route and keep progressing.

Postcondition: participation does not unlock or block a required child step. Mother has not obtained hidden access to answers/reflections.

UC-06 - Author, review, and publish new content

Actor: Admin. **Requirements:** FR-17-FR-21, FR-24.

Preconditions: admin authorization, supported templates known; no learner-data access implied.

1. Admin creates a door/journey or a draft revision of an existing journey.
2. Admin assigns child/mother/shared paths, stable step IDs, ordering, template data, media, and completion rules.
3. Media is uploaded privately; processing and rights/accessibility metadata are checked.
4. Admin previews audience-specific paths and records editorial review, including sources where applicable.
5. Validator checks schema, assets, references, reachable required steps, optionality rules, and supported client capabilities.
6. Admin explicitly publishes using the expected current draft version.
7. Service atomically freezes the revision, changes the live pointer, and records an audit event.
8. Existing compatible clients see the new journey on the next catalog refresh; already open sessions are not forcibly navigated away.

Alternatives/failures: invalid data/unreviewed material to actionable errors; stale draft version to reload/reconcile; publication failure to previous live version remains; upload alone to remains private; same publish operation retried to one published revision and one effective publication action.

Postcondition: a complete revision is live. No app update is needed for content using already supported templates; no arbitrary game-engine creation is implied.

UC-07 - Revise or withdraw content

Actor: Admin. **Requirements:** FR-21-FR-24.

Routine revision: clone live revision to a new draft to review/validate to publish. Existing attempts remain bound to the older available revision. New attempts use the current one. Old completion remains historical.

Withdrawal: admin chooses an exact revision to provides reason to confirms scope to service marks it withdrawn and invalidates new delivery access. Bound attempts receive an unavailable state on their next authorization/content check. Their records are not erased.

Alternatives: no replacement available to honest unavailable state, no fake next journey; correction to withdrawn content to new reviewed revision; previous safe version desired to create a new draft from it and review/publish, not an unreviewed silent rollback.

Limit: already downloaded bytes cannot be guaranteed to disappear immediately from disconnected devices. The offline policy remains under review.

UC-08 - Revoke device access or erase a profile

Actor: Mother. **Requirements:** FR-04, FR-27.

Revoke/replace: mother selects the relevant device to confirms revocation to service invalidates it to protected requests fail to replacement device is paired using UC-01 and restores server-confirmed progress.

Erase child profile: mother sees what will be removed to explicitly confirms to sessions are revoked to profile and learner records enter the agreed deletion process to local clients clear scoped caches on receiving revocation/deletion or logout.

Erase mother account: explain inclusion of all owned child profiles under P-03; require confirmation; revoke all affected sessions and schedule linked-data removal.

Failure: unrelated-account request to reject without revealing profile details. Backup purge and retention timing are unresolved release decisions; do not promise immediate erasure from every backup.

Postcondition: a lost device cannot make new authorized server requests. Revocation is not equivalent to deletion; historical progress survives revocation but is removed/anonymized according to confirmed erasure policy when deletion is requested.

05 Logical architecture

1. Direction

Start with one backend application organized into clear modules, a transactional persistence boundary, and separate media storage/delivery. These are logical responsibilities, not a recommendation to deploy separate microservices. The admin web client is a proposal. Backend framework, identity provider, database, storage/CDN, queues, hosting, and observability products remain undecided.

The content model is the key architectural choice: administrators publish data that existing client renderers understand. This supports new journeys without turning the MVP into a general game-creation platform.

2. Responsibility boundaries

Component	Owns	Does not own
Mobile role shell	Mother/child navigation and accessibility	Server authorization decisions.
Content renderer	Supported template display, authored choices, audio/text controls	Executing remote scripts or inventing new game mechanics.
Local progress store and outbox	Durable checkpoints, pending events, restart recovery	Claiming server sync before acknowledgment.
Identity/family module	Adult identity mapping, ownership, pairing, child sessions, revocation/deletion initiation	Editorial publication.
Catalog/content module	Drafts, immutable revisions, validation, availability, publication pointer	Treating a new schema as supported by old apps.
Progress module	Version-bound attempts, idempotent events, required-step completion	Moral scores or raw reflection storage.
Mother-summary projection	Allowed child topic/completion summary	Pass-through access to detailed learner event data.
Media module	Private upload, type validation, asset readiness, delivery authorization	Automatic publication on upload.
Admin client	Editing, review, preview, explicit publication	Direct database writes or learner-data browsing.
Audit/operations	Content lifecycle audit, non-sensitive faults, backup/restore	Child-answer analytics.

3. Proposed trust boundaries

All clients are untrusted. Server operations derive ownership from validated identity/session scope; passing a child ID never grants access. Preview media is private to authorized admins. Published media is available only according to its revision's delivery policy. Catalog filtering is not sufficient access control for direct asset or revision requests.

A pairing challenge is not itself a permanent credential. A pending device must possess its own pending-request credential, and the owning mother must confirm before issuance. Pending requests expire; approval, redemption,

and consumption must be atomic and duplicate-safe. The precise cryptographic protocol belongs in the detailed auth design, not this conceptual UML.

Content admins manage content only. A future support capability requires an explicit permission and purpose; it must not be silently added to “admin”.

4. Content contract

Each revision has a schema version and declares required client capabilities. Each step has a stable ID within that revision, audience path, template type, authored data, optional media references, and a completion rule. Choice destinations stay inside their authored path unless an explicit supported transition exists.

Publish validation checks:

- Assets exist, are ready, and have the required alternatives/source/rights metadata.
- Required digital steps are reachable; optional tasks do not become hidden gates.
- All references and choice targets resolve; authored loops have an exit.
- Template/schema capabilities are compatible with the intended supported clients.
- Review is recorded and the expected draft version matches the saved version.

Published data is immutable. The active catalog points to a live revision; previous nonwithdrawn revisions remain available for bound attempts. Withdrawal changes availability, not historical content or completion. A content fix requires another reviewed revision.

5. Progress and synchronization

Use distinct learner identities for mother and child. An attempt binds one learner to one content revision. Persist a resume cursor separately from the set of completed required steps.

Proposed event envelope: eventId, attemptId, revisionId, deviceSessionId, sequence, checkpointStepId, and newly completed allowed step IDs. Exclude raw answer/reflection payloads. The server validates step IDs and state transitions; completion measures participation only, so clients are not proving real-world conduct.

Local storage commits before showing “saved on this device”. The outbox retries with the same identifier. The server atomically deduplicates and commits before returning “synced”. Same identifier with a changed payload is an error. Stale cursor updates cannot overwrite a newer cursor; completed steps merge monotonically.

The draft assumes one actively used child device session per child for learning; replacement revokes the previous learning session. This reduces concurrent-cursor ambiguity. If multiple simultaneous child devices are required, add an explicit conflict policy before implementation. Multiple children on their own devices remain supported under P-03.

Full offline playback is not promised. Cached scenes may be usable during a disruption; missing media requires connectivity. A replacement device only restores server-confirmed state. Optional reflection choices are stored separately from the sync outbox and cleared with profile/session removal.

6. Publication consistency

A publish operation uses an idempotency identifier plus expected draft version. Validate before mutation; commit immutable revision, live pointer, and audit record in one transactional boundary. External media must be ready before the transaction. A failure leaves the prior live pointer unchanged. Asset cleanup after abandoned drafts needs a later retention policy.

For withdrawal, deny new authorized delivery and invalidate cached metadata where feasible. Time-bounded media links can reduce exposure, but exact link lifetime/CDN policy is undecided. Do not claim that previously delivered files can be retracted instantly.

7. Data and API design still to come

After the behavioral baseline is accepted, produce:

- Physical schema with keys, constraints, indexes, deletion/retention behavior, and migrations.
- API contracts for accounts/pairing, catalog/revisions, progress, summaries, and admin workflows.
- Concrete auth threat model, credential lifetime/rotation/recovery, abuse controls, and ownership tests.
- Media format/size limits, processing pipeline, caching and storage budget.
- Deployment topology, environment separation, CI/CD, backups, restore targets, monitoring, and cost estimates.

These are not omitted implementation work: the current stage establishes requirements and UML before committing to a stack. Picking providers before OQ-04/OQ-05/OQ-06/OQ-10 would conceal assumptions about platform, connectivity, distribution, and scale.

8. Main risks and proposed controls

Risk	Proposed control	Decision still needed
Pairing the wrong device	Expiry, single pending redemption, mother confirmation, revocation	Accept onboarding mechanism.
Leaking one family's records	Server ownership checks and explicit response projections	Confirm summary visibility.
A content edit breaks an ongoing journey	Immutable revisions and version-bound attempts	Withdrawal/replacement UX.
Admin publishes unsupported interactions	Capability validation and bounded templates	Final template list.
User assumes unsynced work is backed up	Visible local/pending/confirmed state	Connectivity scope.
Production content is missing or unsuitable	Editorial ownership, provenance, preview, age testing	Content team and review process.
Broad scope overwhelms one-topic MVP	One complete vertical journey plus reusable authoring	Approve the proposed MVP boundary.

06 UML models

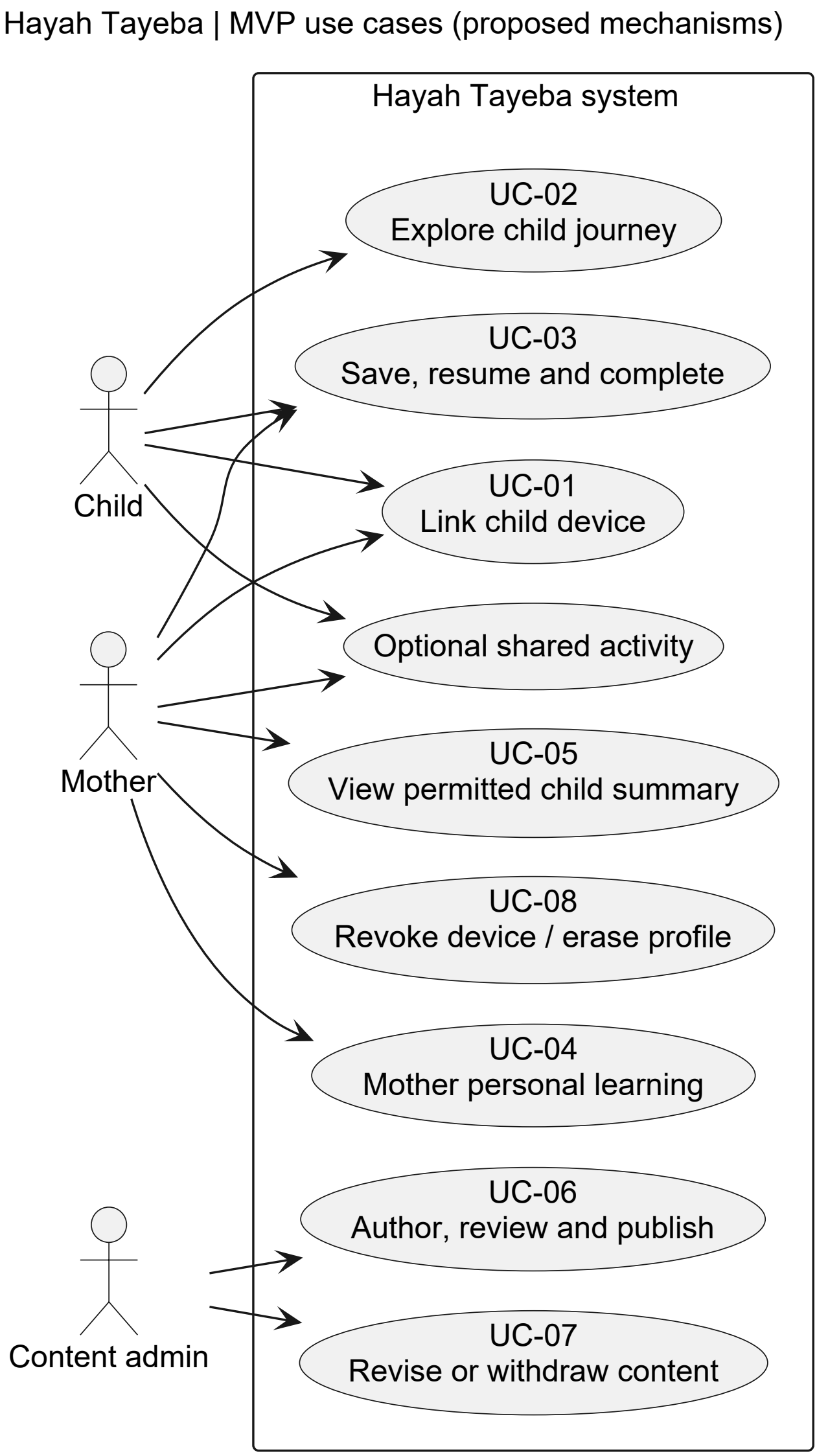
The following eight models express the draft requirements. Behavioral mechanisms, privacy boundaries, and account ownership marked as proposals remain subject to review. These are software models, not UI layouts.

Model	Purpose
Use cases	Three human actors and the system goals. Initial linking involves the mother; learning participation does not depend on her availability. Content administration grants no access to learner records.
Content domain	Conceptual classes, not physical database tables. Stable journeys have immutable revisions with audience-specific paths. Multiplicities describe a valid publishable model; drafts may be incomplete.
Learner domain	Mother and child are separate learners. Attempts bind to a content revision. The proposed ownership and active-device restrictions remain review decisions.
Child learning activity	Independent learning, durable local progress, optional shared/reflection steps, and network recovery. A locally finished session can still have synchronization pending.
Device pairing sequence	Proposed temporary-code flow with explicit mother confirmation before granting child-scoped device access. Detailed credentials and identity-provider selection remain open.
Progress synchronization sequence	Separates local save from server confirmation and handles duplicate submissions. The mother receives an authorized summary rather than raw choices or reflections.
Content revision lifecycle	Publication freezes a revision. Editing creates another draft; withdrawal affects future availability without silently removing historical completion.
Logical components	Proposed client, backend, persistence, identity, and media boundaries. Backend modules do not imply separately deployed microservices or a selected vendor.

Drawing pages use A3 dimensions so the original diagrams remain legible when zoomed or printed at their native size. Editable PlantUML sources are maintained with the project specification.

UML 01 Use cases

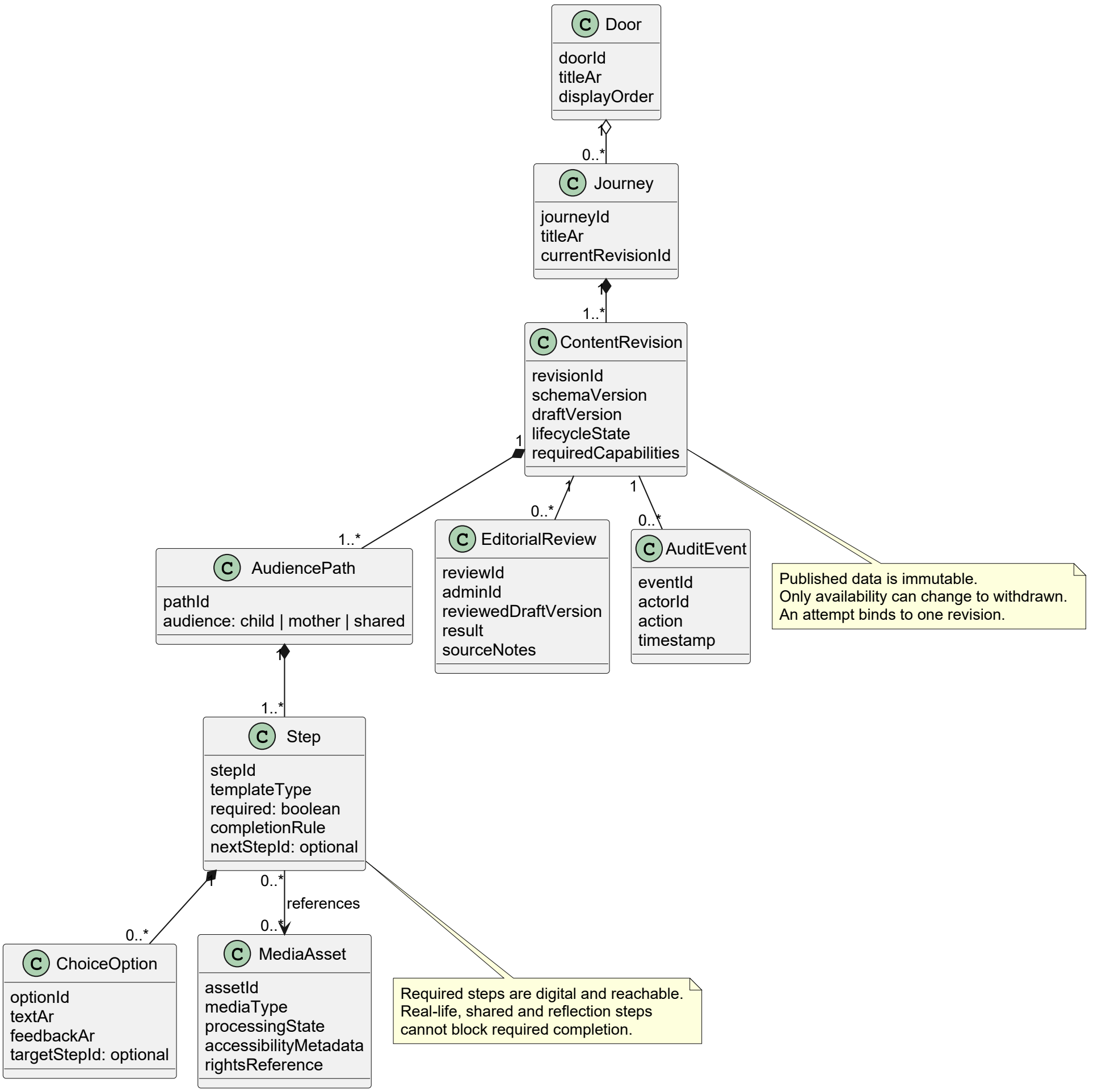
Three human actors and the system goals. Initial linking involves the mother; learning participation does not depend on her availability. Content administration grants no access to learner records.



UML 02 Content domain

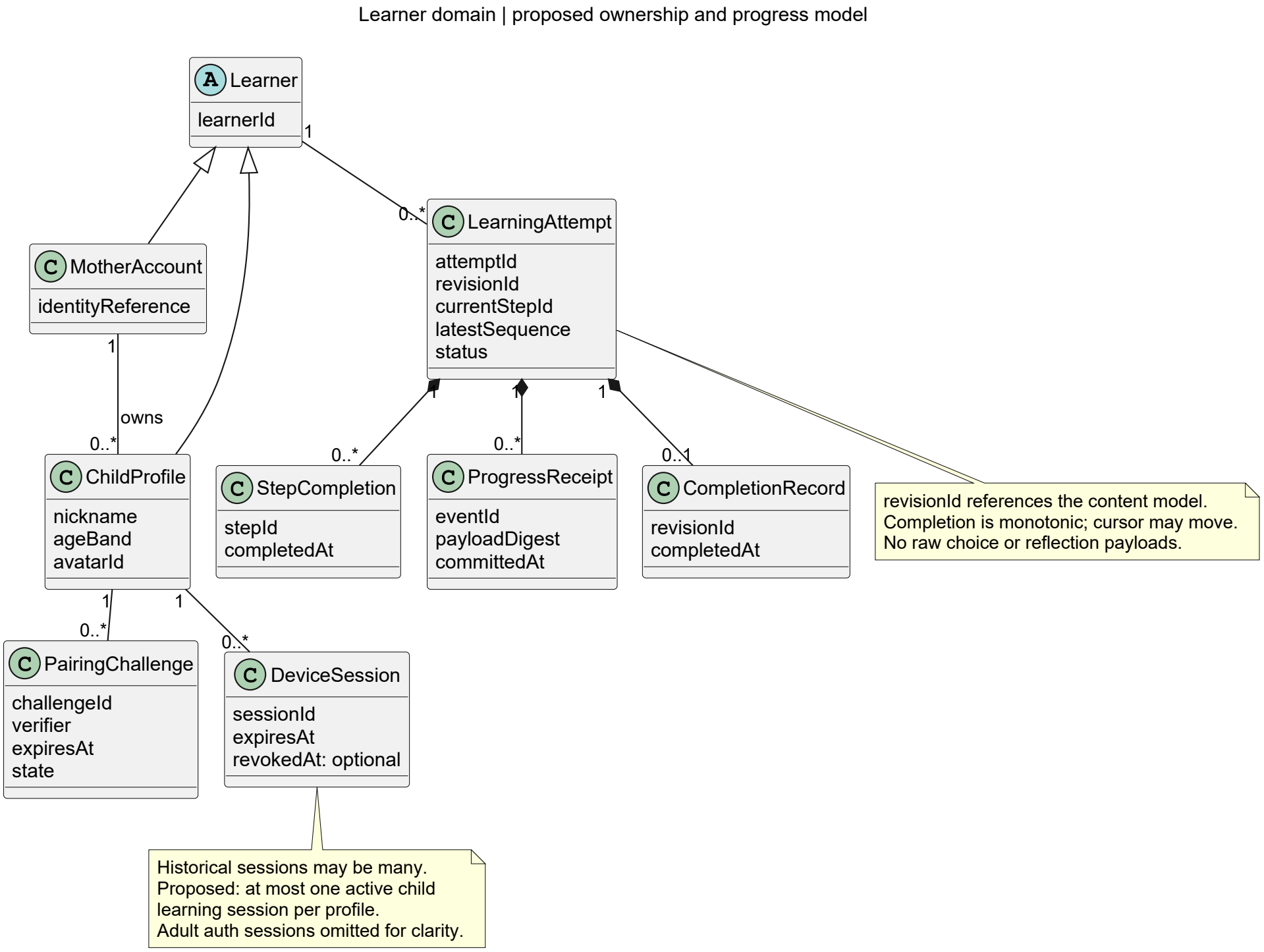
Conceptual classes, not physical database tables. Stable journeys have immutable revisions with audience-specific paths. Multiplicities describe a valid publishable model; drafts may be incomplete.

Content domain | conceptual classes, not database tables



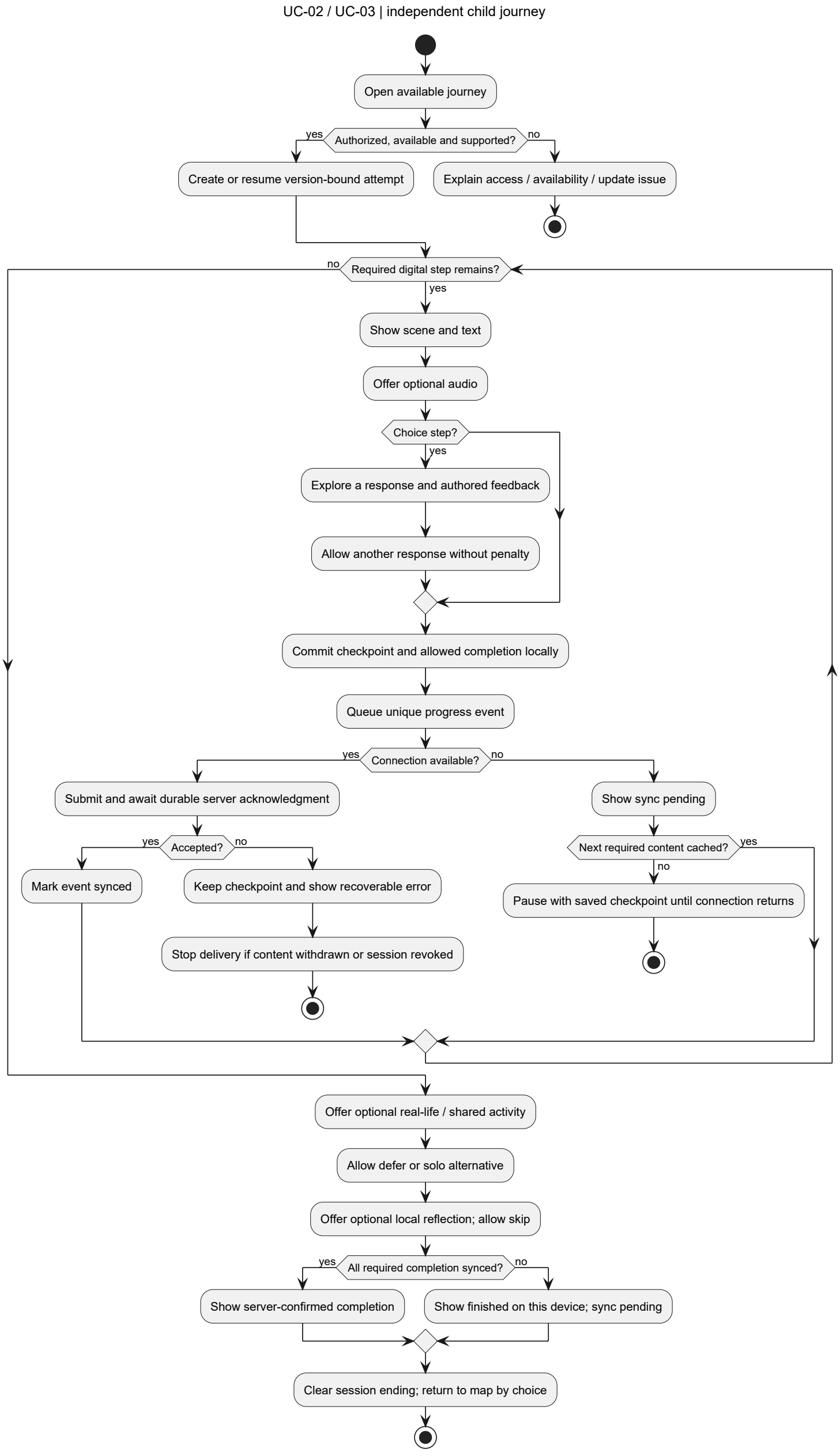
UML 03 Learner domain

Mother and child are separate learners. Attempts bind to a content revision. The proposed ownership and active-device restrictions remain review decisions.



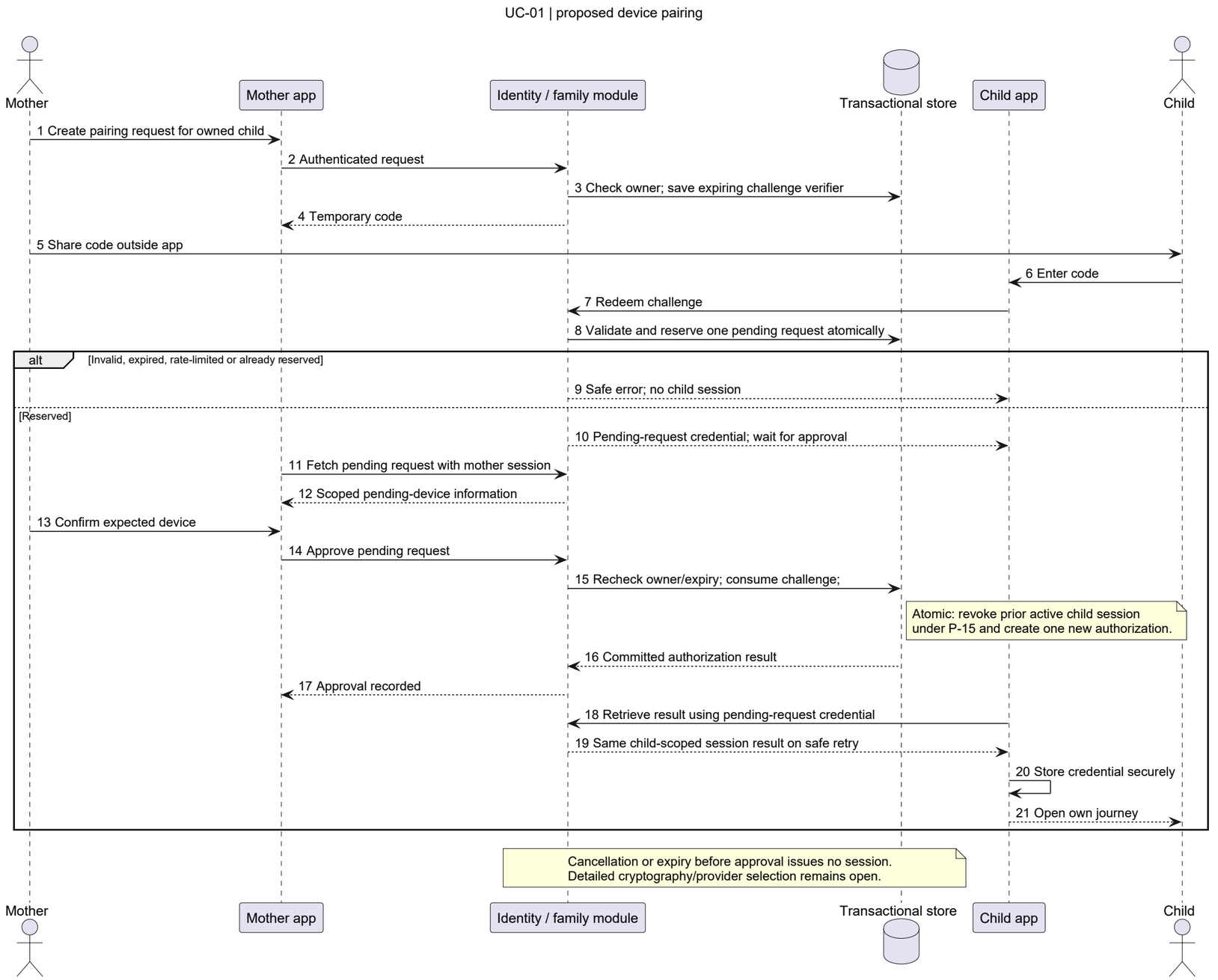
UML 04 Child learning activity

Independent learning, durable local progress, optional shared/reflection steps, and network recovery. A locally finished session can still have synchronization pending.



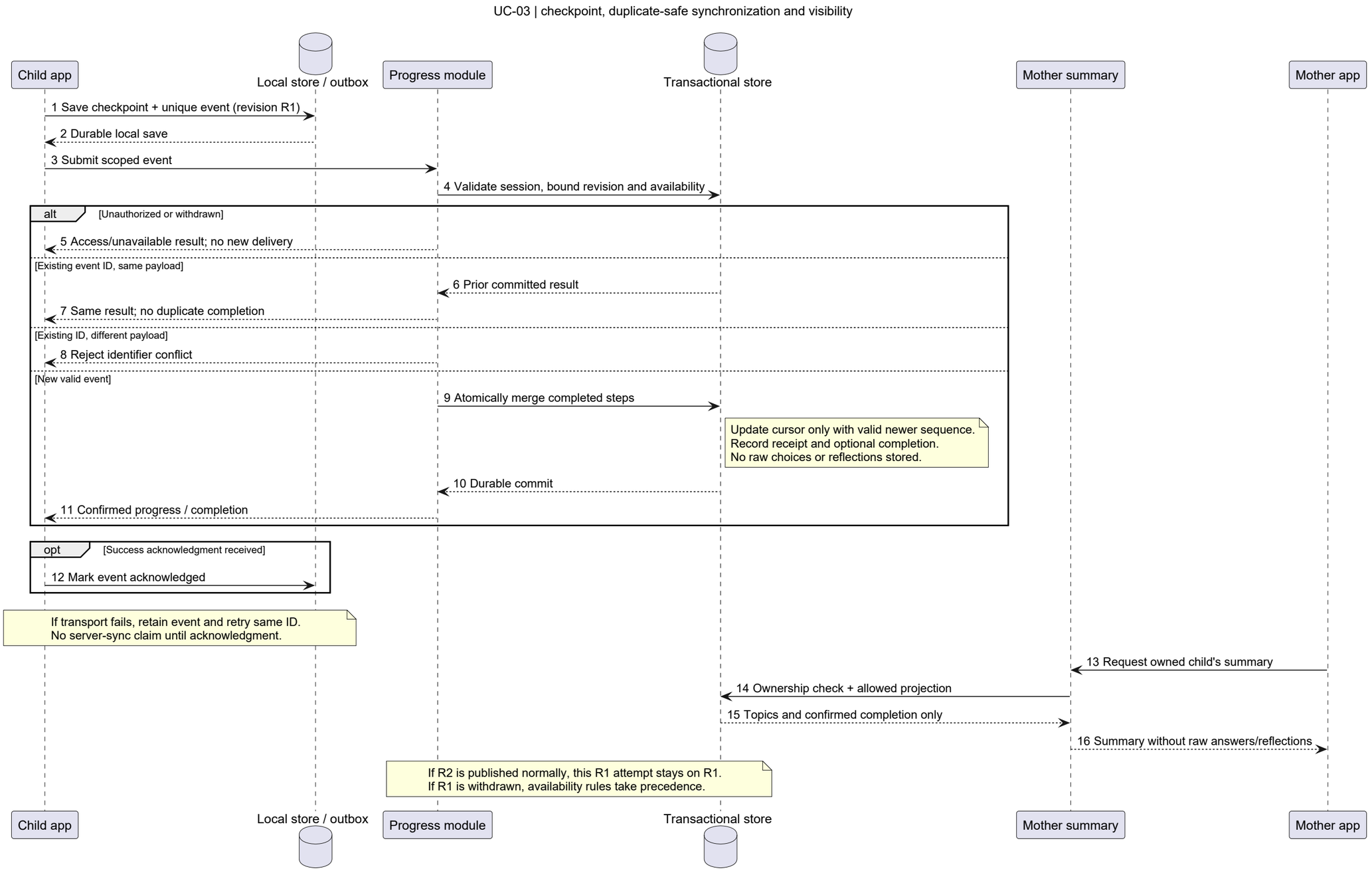
UML 05 Device pairing sequence

Proposed temporary-code flow with explicit mother confirmation before granting child-scoped device access. Detailed credentials and identity-provider selection remain open.



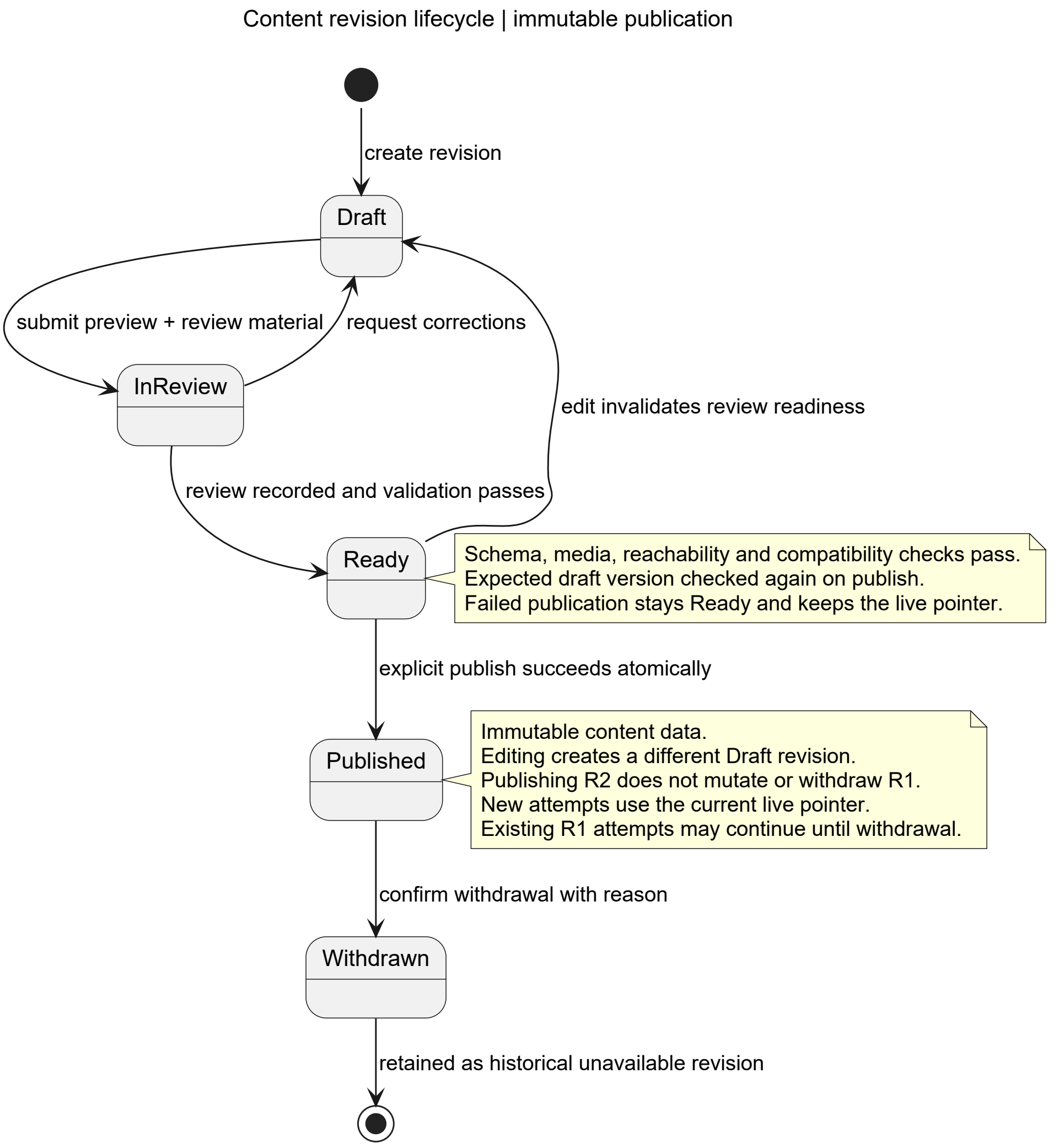
UML 06 Progress synchronization sequence

Separates local save from server confirmation and handles duplicate submissions. The mother receives an authorized summary rather than raw choices or reflections.



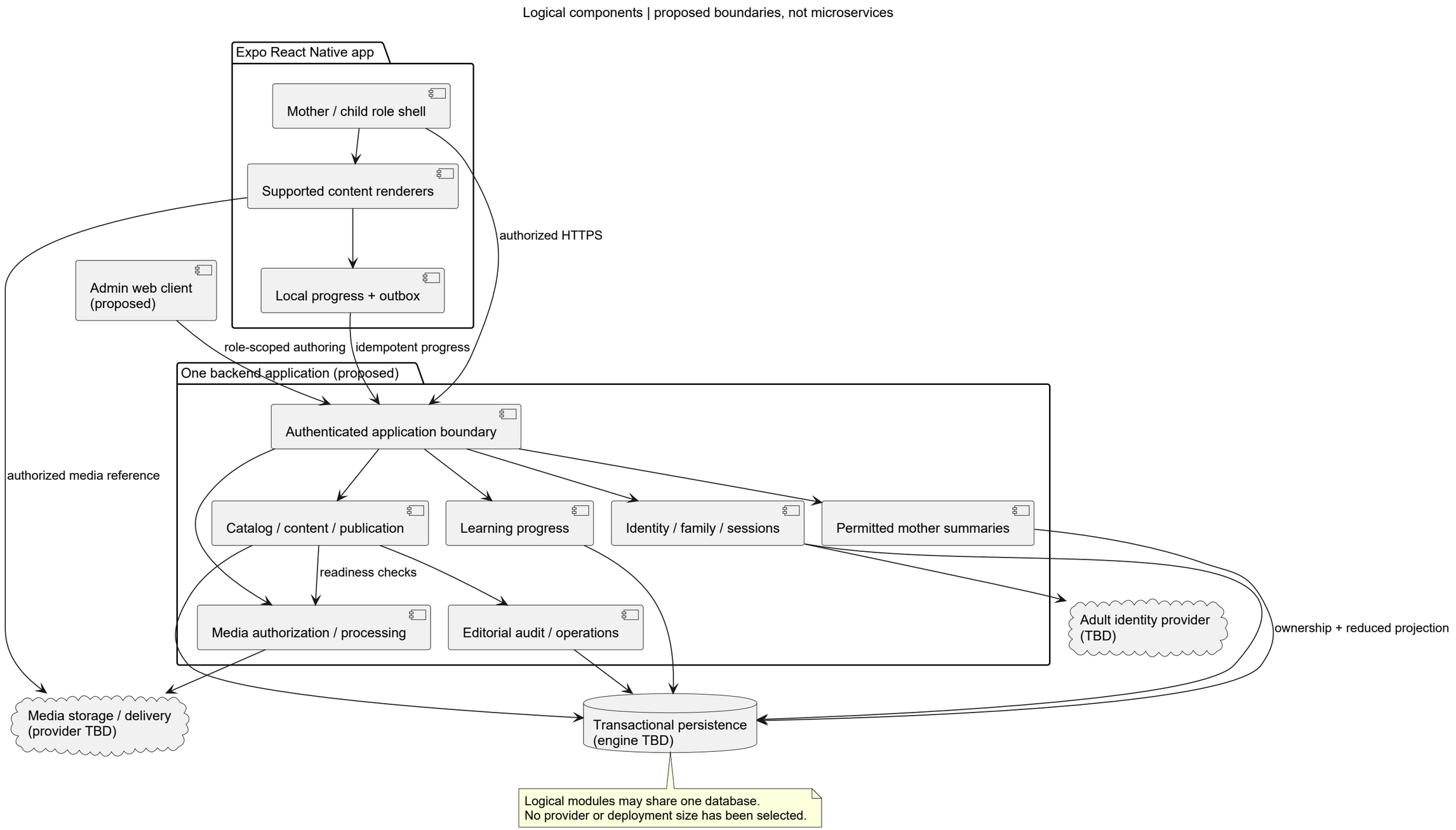
UML 07 Content revision lifecycle

Publication freezes a revision. Editing creates another draft; withdrawal affects future availability without silently removing historical completion.



UML 08 Logical components

Proposed client, backend, persistence, identity, and media boundaries. Backend modules do not imply separately deployed microservices or a selected vendor.



07 Requirements traceability

These are planned acceptance checks. They have not been executed against an application.

Model keys: U1 use cases; U2 content classes; U3 learner classes; U4 child activity; U5 pairing sequence; U6 progress sequence; U7 content lifecycle; U8 components. See the numbered models in Section 06.

Functional coverage

Requirement	Use case	Models	Planned acceptance check
FR-01	UC-01, UC-06	U1, U5, U8	AT-01: try anonymous, wrong-role, and unrelated-family requests; deny each protected operation.
FR-02	UC-01	U3, U5	AT-02: create a valid child profile without contact data; reject unsupported age and cross-owner binding.
FR-03	UC-01	U5	AT-03: happy-path pairing; expired/cancelled/reused code; concurrent redemption; wrong-owner approval; retry lost issuance response.
FR-04	UC-08	U3, U5	AT-04: revoke old device; reject its next protected request; replacement restores only server-confirmed progress.
FR-05	UC-01, UC-04	U3, U8	AT-05: mother and child completion remain separate; child cannot enter mother's account context.
FR-06	UC-02	U1, U4	AT-06: one-door catalog works; draft and withdrawn entries do not appear as startable content.
FR-07	UC-02	U2, U4, U8	AT-07: render supported manifest; safely reject future schema/template without corrupting an existing attempt.
FR-08	UC-02	U2, U4	AT-08: read/listen/navigation with audio off, unavailable, or interrupted; essential meaning remains accessible.
FR-09	UC-02	U2, U4	AT-09: every authored choice resolves; try another response; no moral score or completion reset.
FR-10	UC-02, UC-05	U1, U4	AT-10: complete all required learning with no mother participation after setup.
FR-11	UC-02	U2, U4	AT-11: defer real-life task and skip reflection; required progression still works.
FR-12	UC-03	U3, U6	AT-12: save, kill app, resume, lose network, retry duplicate event, reject changed-payload ID reuse.
FR-13	UC-03	U3, U4, U6	AT-13: required-step set yields one completion per attempt; optional skips/replay preserve history.
FR-14	UC-04	U1, U2, U3	AT-14: mother starts and completes her own reviewed content regardless of child state.

Requirement	Use case	Models	Planned acceptance check
FR-15	UC-05	U6, U8	AT-15: inspect parent-summary response; only linked child's permitted fields appear.
FR-16	UC-05	U1, U4	AT-16: shared activity can be declined or done asynchronously; solo route remains available.
FR-17	UC-06	U1, U2, U8	AT-17: publish a second supported journey; unchanged client binary retrieves it.
FR-18	UC-06	U2	AT-18: validate every supported template, malformed step, broken target, and optional-step gate.
FR-19	UC-06	U2, U8	AT-19: incomplete/invalid upload cannot publish; draft media is inaccessible to child session.
FR-20	UC-06	U2, U7	AT-20: preview paths and require review; fail on missing alternatives, bad sources/references, unreachable steps.
FR-21	UC-06, UC-07	U2, U7	AT-21: failed, repeated, or stale publish leaves an atomic live pointer and no partial revision.
FR-22	UC-03, UC-07	U2, U3, U6	AT-22: publish R2 during an R1 attempt; old attempt stays on R1 and historical completion survives.
FR-23	UC-07	U7, U8	AT-23: withdraw bound revision; block subsequent online delivery, preserve records, show unavailable state.
FR-24	UC-06, UC-07	U2, U8	AT-24: verify actor/version/action/outcome audit records without credentials or child answers.
FR-25	UC-02, UC-03	U4, U6	AT-25: fail required/optional media and network separately; maintain checkpoint and meaningful retry.
FR-26	UC-03, UC-04	U3, U6	AT-26: choose/skip reflection, inspect requests/logs; no reflection leaves device; profile removal clears it.
FR-27	UC-08	U3, U8	AT-27: confirmed erasure request revokes scope immediately; unrelated requester denied; agreed deletion process verified.

Quality coverage

Requirement	Planned check	Baseline dependency
NFR-01	AT-N01: object-level authorization matrix across account, progress, summary, content, and media operations.	Final API surface.
NFR-02	AT-N02: expired/replayed pairing, atomic redemption, rate limit, secure credential storage, token-free logs.	Approved auth mechanism and abuse policy.
NFR-03	AT-N03: network/telemetry/log inspection and deletion/retention exercise.	Retention and release-country decisions.

Requirement	Planned check	Baseline dependency
NFR-04	AT-N04: local restart, server transaction failure, reordered events, duplicate submissions.	Concrete persistence implementation.
NFR-05	AT-N05: measure p95 local interaction timing against proposed 200 ms target.	Agreed representative device.
NFR-06	AT-N06: measure p95 metadata/progress latency against proposed 1 second target.	Agreed concurrency, network, region, and dataset.
NFR-07	AT-N07: screen reader, 200% text, RTL, captions, non-color feedback, reduced motion.	Supported OS/device matrix.
NFR-08	AT-N08: release-build smoke and regression tests on supported platforms and admin browsers.	Launch platform/version decision.
NFR-09	AT-N09: old supported client versus incompatible manifest; safe fallback and preserved progress.	Compatibility policy and template versions.
NFR-10	AT-N10: injected sync/publish faults, correlation IDs, backup restore rehearsal.	Owner, recovery targets, backup/retention policy.
NFR-11	AT-N11: source/rights and age-suitability review exists for every published revision.	Appointed editorial owner and completed content.
NFR-12	AT-N12: observed child usability sessions across the age range.	Sample/recruitment plan and agreed threshold.

Confirmed product needs mapped to the specification

Product decision	Coverage
C-01: ages 8-12	FR-02, NFR-11, NFR-12
C-02: mother's own journey and support	FR-05, FR-14-FR-16
C-03: child independence / optional mother	FR-10, FR-11, FR-16, BR-01
C-04: chosen first journey	SRS scope, FR-06, UC-02
C-05: light fantasy / everyday dialogue	FR-06, UC-02, editorial review
C-06: Expo React Native	SRS interfaces, NFR-08, architecture / U8
C-07: separate devices	FR-03-FR-05, FR-12, U5, U6
C-08: admin content updates	FR-17-FR-24, UC-06/UC-07
C-09: one-topic MVP	SRS scope, FR-06, AT-17 tests later extensibility

Product decision	Coverage
C-10: presentation only	Executive brief and source limitations; production content remains a dependency
C-11: documents before UI	This pack; no UI implementation or visual design is performed

Documentation verification versus product verification

Documentation checks validate file links, unique IDs, coverage, and renderable diagram syntax. They cannot prove that an implementation is secure, a child understands the experience, the religious content is correct, or the service meets latency targets. Those checks occur after the relevant content and software exist.

08 Review and document control

Review sequence

1. Confirm the first-release scope and the proposed family/account model.
2. Agree mother visibility, child privacy, and progress expectations across devices.
3. Approve supported content templates and assign editorial ownership.
4. Resolve launch-platform, connectivity, operating, and data-retention decisions.
5. Baseline the SRS, then update the use cases, UML, and acceptance mapping together.

Change control

Keep requirement IDs stable. Record an explicit decision before promoting a proposal to the approved baseline. Revise related diagrams and acceptance checks when behavior changes. A routine content edit must not silently rewrite historical learner completion.

Source material

- Presentation 13.pptx: supplied conceptual project presentation.
- Expanded project notes: supplied written material about the learning themes and journey.
- Product-planning decisions: confirmed audience, mother/child participation model, first topic, visual direction, framework, separate devices, and admin content updates.

Verification status

The source pack contains 27 functional requirements, 12 nonfunctional requirements, 11 business rules, eight detailed use cases, and eight UML diagrams. Requirement identifiers and traceability coverage were checked, and the diagrams were rendered from their editable sources. These documentation checks do not substitute for application, security, accessibility, content, or usability acceptance testing.

Review record

Record agreed decisions and revisions here before establishing an approved baseline.

Field	Review entry
Document version	0.1 - client review draft
Review date	To be recorded
Reviewer / role	To be recorded
Decisions accepted	Reference C-, P-, OQ-, or requirement IDs
Changes requested	Reference requirement IDs and affected models
Baseline status	Not yet approved